Politecnico di Milano

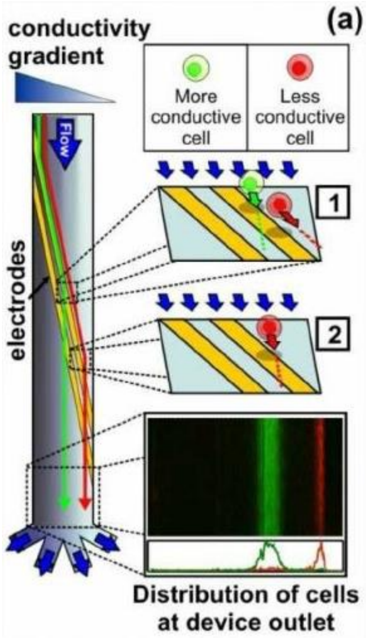
Biochip A.Y. 2019/2020 - M. Carminati

On-Line Exam of 27.08.2020

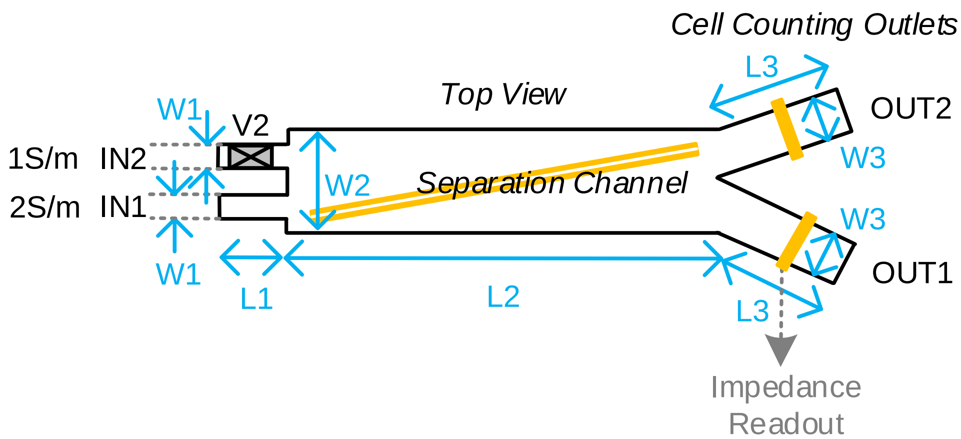
Test duration: 2 hours – No support material admitted.

All the answers must be motivated. Beyond the final numerical results, all the expressions must be provided.

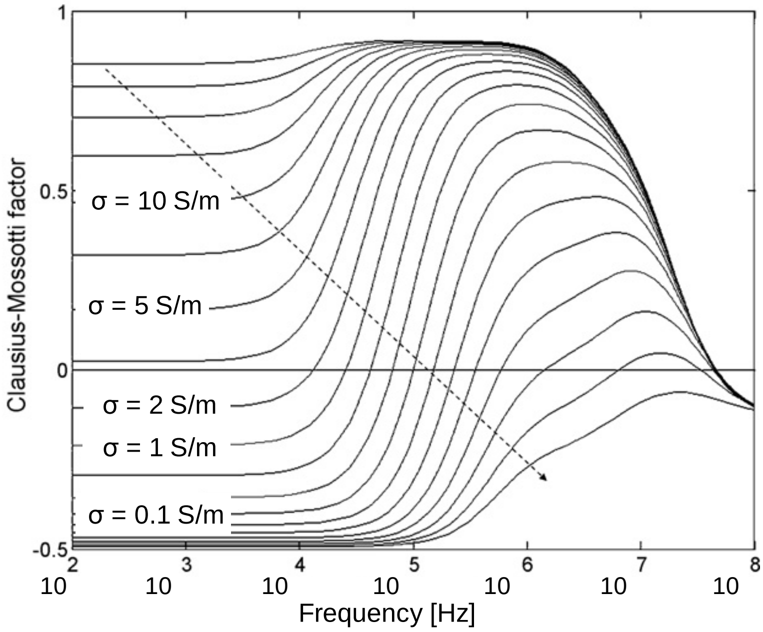
Consider the following microfluidic system used for separation of cells based on their electrical properties, through a different iso-dielectric equilibrium point. A transverse conductivity gradient in the solution is created across the separation channel from 2 to 1 S/m. All cells enter from IN1 (max. conductivity) and experience repulsion from a DEP barrier created by oblique electrodes running at the bottom of the channel. Sliding along the barrier, they move into regions with lower conductivity, until they reach an equilibrium point where the DEP force is zero and thus they can pass the barrier, proceeding straight ahead. The equilibrium point is different for different cell types, thus being spatially separated in the direction orthogonal to the flow. Impedance counters are used on the outlets to count the number of passing cells.



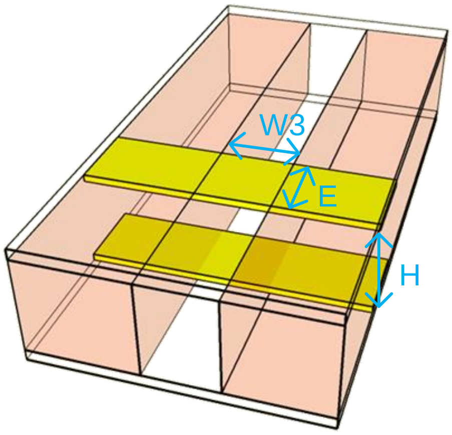
1. Consider the following planar dimensions of the device: L1=5mm, L2=50mm, L3=10mm, W1=50µm, W2=4mm, W3=2mm. The height is constant in the whole device and equal to H=100µm. Find the electrical equivalent of the fluidic circuit when valve V2 (used to create a conductivity gradient) is closed.



2. If $P_{OUT1} = P_{OUT2}$ and Q_{IN}=1µL/min, find the velocity in each section of the device (V2 closed).
3. Assuming that the fluid velocity in section L2 is equal to 50mm/s and that the diffusion distance in the direction orthogonal to the flow is equal to W2/2=2mm at the end of the chamber, find the diffusion coefficient of the salt molecules.
4. Discuss if the DEP force generated by the barriers must be negative or positive. Discuss also the role of the cell dimensions on the DEP force.
5. Considering the following plot of the real part of Clausius-Mossotti factors, find a proper frequency for the AC field that ensure a repelling barrier for cells entering in IN1.



6. Describe the fabrication steps required to pattern gold electrodes on a glass substrate.
7. Considering the geometry of electrodes (with E=1mm) in the figure, find and plot the equivalent impedance in both output channels, assuming a liquid conductivity of 2S/m in the whole OUT1 and 1S/m in OUT2. Assume a parasitic capacitance between the electrodes due to the connections of about 10pF.



8. Given the spectra found above, design a circuit to read the impedance on each outlet (properly size all components)
9. Choose the optimal measuring frequencies for both outlets in order to count cells.
10. Replacing the electrodes impedance with a capacitor of 1nF at the input of the current amplifier, compute and plot the input-referred total noise spectrum, assuming en=1nV/sqrt(Hz) and in=10fA/sqrt(Hz) for the opamp.
11. Choose the lock-in low-pass filter bandwidth to achieve a maximum counting rate of 1000 cells/sec.
12. Propose a solution to cope with variations of the temperature, affecting the liquid conductivity and thus, the counting.